MIL-C-3098/71B (USAF)
11 April 1975
SUPERSEDING
MIL-C-3098/71A (USAF)
9 October 1968

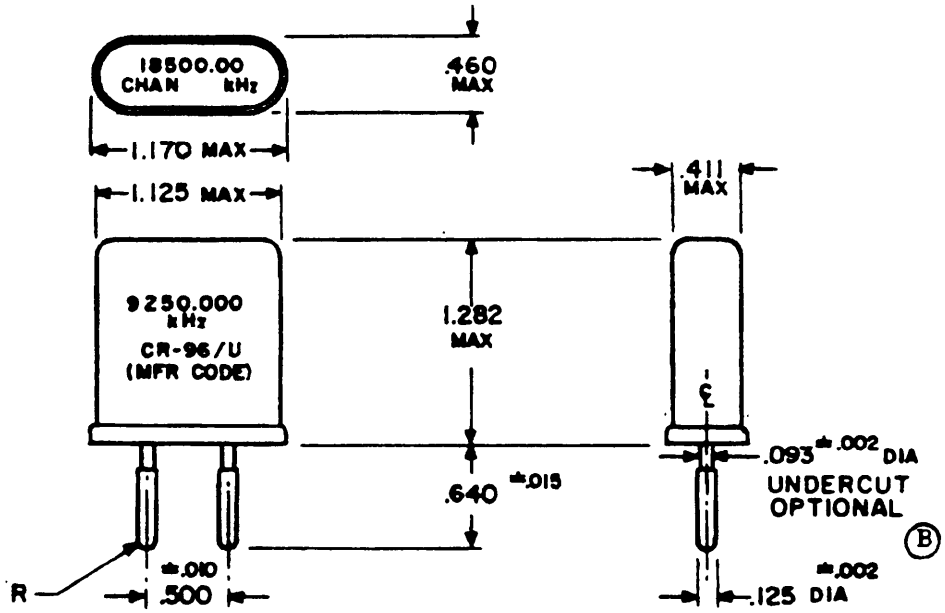
MILITARY SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR-96/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the crystal unit described herein shall consist of this document and the latest issue of Military Specification MIL-C-3098.

Pertinent characteristics: 2 to 9.25 MHz; fundamental; noncontrolled; antiresonance.



INCHES	MM	INCHES	MM	INCHES	MM
.002	.05	.125	3.18	.640	16.26
.010	.25	.411	10.44	1.125	28.58
.015	.38	.460	11.68	1.170	29.72
.093	2.36	.500	12.70	1.282	32.56

NOTES:-
1. DIMENSIONS ARE IN INCHES.
2. METRIC EQUIVALENTS (TO THE NEAREST .01 mm) ARE GIVEN FOR GENERAL INFORMATION ONLY AND ARE BASED UPON 1 INCH = 25.4 mm.

Ⓑ DENOTES CHANGES

FSC 5955

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TABLE I. Requirements

Requirement	Specified Value or Condition
Holder	HC-11A/U (MIL-H-10056/17)
(E) Dimensions, marking, and configuration	See Figure
Frequency range	2.0 to 9.25 MHz, inclusive
Mode of oscillation	Fundamental
Test set, calibration values, and rated drive level	See Table II
ntiresonance, load capacitance	35.0 ±0.5 pf
Operating temperature range (noncontrolled)	-35° to +75°C, inclusive
Frequency tolerance	±0.005 percent
Equivalent resistance	See Table III
Shock:	
Frequency change permitted	±0.0005 percent
Equivalent resistance change permitted	±15 percent, or 2 ohms, whichever is greater
Vibration: 2/	
Frequency change permitted	±0.0005 percent
Equivalent resistance change permitted	±15 percent, or 2 ohms, whichever is greater
Temperature cycling:	
Frequency change permitted	±0.0005 percent
Equivalent resistance change permitted	±15 percent, or 2 ohms, whichever is greater
Aging:	
Frequency change permitted	0.0005 percent

1/ Marking Information - Procurement documents shall specify the following:

a. The channel (operating) frequency shall be stamped on top of the holder in kilohertz (kHz).

b. The crystal frequency shall be stamped on side of the holder in kilohertz (kHz).

2/ Method 201 of MIL-STD-202.

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TABLE II. Drive Level Calibration Data

Government Reference Standard Test Set	Band of Specified Frequencies, Inclusive	Calibration Values			Rated Drive Level
		Resistance	Crystal Current	Calibrating Frequency	
	MHz	Ohms	ma	MHz	mw
TS-330/TSM	2.0 to 2.26	50	15	2.0	
	2.26+ to 3.41	40	15	3.0	
	3.41+ to 5.11	25	10	Crystal Frequency	10.0 ±2.0
	5.11+ to 7.5	14	25	Crystal Frequency	10.0 ±2.0
	7.5+ to 9.25	11	30	Crystal Frequency	10.0 ±2.0

TABLE III. Equivalent-Resistance

Frequency range, inclusive	Maximum resistance
MHz	Ohms
2.0 to 4.75	120
4.75+ to 6.0	75
6.0+ to 7.5	50
7.5+ to 9.25	35

NOTES:

A. This crystal-unit type is to be used for special applications, and is not for general use. Specific approval of the Government is required prior to each intended use.

B. These crystal units are for use in the Collins 18S Transmitter. They were previously fabricated to the requirements specified in Air Force Technical Order 12-1-20 and/or Collins Drawing Number 290-2934/3133-00, 290-7740/7939-00, 290-9728/9927-00, and 291-5451/5950-00.

C. Formulae for computing the crystal frequencies applicable to the Collins 18S Transmitter are:

MHz	
2.0 to 6.0	$F_x = F_o$
6.0+ to 18.5	$F_x = \frac{F_o}{2}$

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Custodians:
Air Force - 84

Review activities:
Air Force - 11, 85

User activities:
Air Force - 17, 19

Preparing activity:
Air Force - 84

Project No. (5955-F171-4)

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OMB Approval
No. 22-R255

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MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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☐ YES ☐ NO *(If "Yes", in what way?)*

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DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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